4-30-26 (WEEK 10)

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EXAMPLE (CORRECTED QUESTION)

Let X, Y have the joint density function

$$f_{X,Y}(x,y) = \begin{cases} e^{-(x+y)}, & \cancel{x+y \geq 0}\ x,y \geq 0 \\ 0, & \cancel{x+y<0}\ x<0 \text{ or } y<0 \end{cases}$$

1. What is $\mathbb{P}(X>Y)$?
2. What is $\mathbb{P}(X \leq t)$, $t \in \mathbb{R}$?
3. What is the distribution of $\frac{X}{Y}$?

Solution

1) $C := \{(x,y) \mid x>y\}$

Then $\mathbb{P}(X>Y) = \mathbb{P}((x,y)\in C) = \iint_C f_{X,Y}(u,v)\,du\,dv = \int_{-\infty}^{\infty}\int_{-\infty}^{u} f_{X,Y}(u,v)\,dv\,du$

$\overset{x,y\geq 0}{=} \int_0^\infty \int_0^u e^{-(u+v)}\,dv\,du = \int_0^\infty e^{-u}\int_0^u e^{-v}\,dv\,du = \int_0^\infty e^{-u}(-e^{-u}+1)\,du = \int_0^\infty e^{-u}-e^{-2u}\,du = \left[-e^{-u}+\frac{1}{2}e^{-2u}\right]_0^\infty = 1-\frac{1}{2} = \frac{1}{2}$

2) Fix some $t \in \mathbb{R}$.

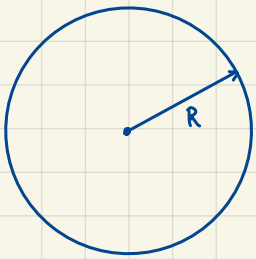
Then, $\mathbb{P}(X \leq t) = \iint_{\{x\leq t\}} f_{X,Y}(u,v)\,du\,dv = \int_0^\infty \int_0^t e^{-(u+v)}\,du\,dv = \int_0^\infty e^{-v}\left[-e^{-u}\right]_0^t dv = \int_0^\infty e^{-v} - e^{-(v+t)}\,dv = \left[-e^{-v}+e^{-(v+t)}\right]_0^\infty = 1-e^{-t}$

3) $F_{\frac{X}{Y}}(t) = \mathbb{P}(\frac{X}{Y} \leq t) = \mathbb{P}(X \leq tY) = \int_0^\infty \int_0^{tv} e^{-(u+v)}\,du\,dv = \int_0^\infty e^{-v}(1-e^{-tv})\,dv = \int_0^\infty e^{-v} - e^{-(t+1)v}\,dv = \left[-e^{-v}+\frac{1}{t+1}e^{-v}\right]_0^\infty = 1-\frac{1}{t+1}$ for $t \geq 0$; 0 for $t<0$

$\therefore f_{\frac{X}{Y}}(t) = F'_{\frac{X}{Y}}(t) = \left(1-\frac{1}{t+1}\right)' = \frac{1}{(t+1)^2}$ for $t \geq 0$; 0 if $t<0$

EXAMPLE

Consider a dart board:



The dart lands at a unif rand point (x,y) inside the dart board.

a) Find the joint PDF of X, Y

b) Find the marginal PDF of X

c) What is the expectation of D (distance from center)

Solution

a) Since (X,Y) is uniform on the board,

$$f_{X,Y}(x,y) = \begin{cases} c, & x^2+y^2 \leq R^2 \\ 0, & x^2+y^2 > R^2 \end{cases}, \quad c: \text{const.}$$

As $1 = \iint_{\mathbb{R}^2} f_{X,Y}(u,v)\,du\,dv = \iint_{\{u^2+v^2\leq R^2\}} c\,du\,dv = c\cdot \pi R^2 \Rightarrow c = \frac{1}{\pi R^2}$

b) Notice, $F_X(t) = \mathbb{P}(X \leq t) = \iint_{\{u\leq t\}} f_{X,Y}(u,v)\,du\,dv$

$$= \begin{cases} 0, & t<-R \text{ or } t>R \\ \underbrace{\iint_{\{u\leq t,\, u^2+v^2\leq R^2\}} f_{X,Y}(u,v)\,du\,dv}_{(*)}, & -R\leq t\leq R \end{cases}$$

Here, $(*) = \int_{-R}^{t}\int_{-\sqrt{R^2-u^2}}^{\sqrt{R^2-u^2}} \frac{1}{\pi R^2}\,dv\,du = \int_{-R}^{t} \frac{2}{\pi R^2}\sqrt{R^2-u^2}\,du$

$\therefore f_X(t) = F'_X(t) = \begin{cases} 0, & -R\leq t\leq R \\ \frac{2}{\pi R^2}\sqrt{R^2-u^2}, & t\notin[-R,R] \end{cases}$ is the marginal PDF

c) $F_D(t) = \mathbb{P}(D\leq t) = \mathbb{P}(x^2+y^2 = D^2 \leq t^2) = \iint_{\{u^2+v^2\leq t^2\}} f_{X,Y}(u,v)\,du\,dv = \frac{\pi t^2}{\pi R^2}$

$\therefore D$ has PDF $f_D(t) = F'_D(t) = \begin{cases} \frac{2t}{R^2}, & 0\leq t\leq R \\ 0, & o/w \end{cases} \Rightarrow \mathbb{E}[D] = \int_{\mathbb{R}} t f_D(t)\,dt = \int_0^R \frac{2t^2}{R^2}\,dt = \left[\frac{2}{3}t^3\cdot\frac{1}{R^2}\right]_0^R = \frac{2}{3}R$

INDEPENDENT RANDOM VARIABLES

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DEFINITION

We say that two random variables X, Y are independent if $\forall A, B \subseteq \mathbb{R}$, we have $\mathbb{P}(X \in A \text{ and } Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$

PROPOSITION

Random variables X, Y are independent $\Leftrightarrow \forall x, y$, $F_{X,Y}(x,y) = F_X(x)F_Y(y)$

Proof

"$\Rightarrow$": $F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \in (-\infty, x], Y \in (-\infty, Y]) \overset{\text{indep}}{=} \mathbb{P}(X \in (-\infty, x])\mathbb{P}(Y \in (-\infty, Y]) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y) = F_X(x)F_Y(y)$

"$\Leftarrow$": We show it's true if $A = (a_1, a_2)$ and $B = (b_1, b_2)$

Then, $\mathbb{P}(X \in A, Y \in B) = F_{X,Y}(a_2, b_2) + F_{X,Y}(a_1, b_1) - F_{X,Y}(a_1, b_2) - F_{X,Y}(a_2, b_1)$

$\overset{\text{assumption}}{=} F_X(a_2)F_Y(b_2) + F_X(a_1)F_Y(b_1) - F_X(a_1)F_Y(b_2) - F_X(a_2)F_Y(b_1)$

$= (F_X(a_2) - F_X(a_1))(F_Y(b_2) - F_Y(b_1))$

$= \mathbb{P}(X \in [a_1, a_2])\mathbb{P}(Y \in [b_1, b_2])$

$= \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ □

REMARK

For discrete random variables, $F_{X,Y}(x,y) = F_X(x)F_Y(y) \Leftrightarrow P_{X,Y}(x,y) = P_X(x)P_Y(y)$

Proof

"$\Leftarrow$": $F_{X,Y}(a,b) = \sum_{x \leq a, y \leq b} P_{X,Y}(x,y) = \sum_{x \leq a, y \leq b} P_X(x)P_Y(y) = \sum_{x \leq a} P_X(x) \sum_{y \leq b} P_Y(y) = F_X(x)F_Y(y)$

"$\Rightarrow$": Similar □

EXAMPLE

Atletico Madrid vs Arsenal

#penalties $\sim Poi(\lambda)$

When a penalty is awarded, it is for:

$\begin{cases} \text{Atletico w.p. } p \\ \text{Arsenal w.p. } 1-p \end{cases}$

indep of all other penalties

Let X = #Atl penalties, Y = #Ars penalties

Then, $P_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y) = \mathbb{P}(\{X=x, Y=y\} \cap \{X+Y=x+y\}) \overset{\text{def}}{=} \mathbb{P}(X=x, Y=y \mid X+Y=x+y)\mathbb{P}(X+Y=x+y)$

Since $X+Y \sim Poi(\lambda)$, thus $\mathbb{P}(X+Y=x+y) = e^{-\lambda} \cdot \frac{\lambda^{x+y}}{(x+y)!}$

Also, as each penalty goes to Atl or Ars indep, if we condition on $x+y$ penalties itself, we get $X \sim Bin(x+y, p)$

Thus, $\mathbb{P}(X=x, Y=y \mid X+Y=x+y) = \mathbb{P}(Y=x \mid X+Y=x+y) = \mathbb{P}(Bin(x+y, p) = x) = \binom{x+y}{x} p^x (1-p)^y$

$\therefore P_{X,Y}(x,y) = \mathbb{P}(X=x, Y=y) = \binom{x+y}{x} p^x (1-p)^y e^{-\lambda} \frac{\lambda^{(x+y)}}{(x+y)!} = \frac{e^{-\lambda} p^x (1-p)^y \lambda^{(x+y)}}{x!\,y!} = e^{-\lambda}\left(\frac{p^x \lambda^x}{x!}\right)\left(\frac{(1-p)^y \lambda^y}{y!}\right) = \left(e^{-\lambda p}\frac{(\lambda p)^x}{x!}\right)\left(e^{-\lambda(1-p)}\frac{(\lambda(1-p))^y}{y!}\right) = P_X(x)P_Y(y)$

$\because P_X(x) = \sum_{y=0}^{\infty} P_{X,Y}(x,y)$, same for $P_Y(y)$

$\therefore X \sim Poi(\lambda p)$, $Y \sim Poi(\lambda(1-p))$ and X, Y: indep

PROPOSITION

If X, Y are cont. rand vars in a probability space $(S, \mathbb{P})$, and their joint PDF factors $f_{X,Y}(x,y) = g(x)h(y)$, then X, Y: indep

Proof

Let $c_1 = \int_{\mathbb{R}} g(x)dx$, $c_2 = \int_{\mathbb{R}} h(y)dy$

Also, $1 = \iint_{\mathbb{R}^2} f_{X,Y}(x,y)dxdy = \int_{\mathbb{R}} g(x)dx \cdot \int_{\mathbb{R}} h(y)dy = c_1c_2$

Notice too: $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)dy = \int_{\mathbb{R}} g(x)h(y)dy = c_2 g(x)$

Similarly, $f_Y(y) = c_1 h(y)$

$\therefore f_{X,Y}(x,y) = g(x)h(y) = 1 \cdot g(x)h(y) = (c_2 g(x))(c_1 h(y)) = f_X(x)f_Y(y)$ □

EXAMPLE

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$f_{X,Y}(x,y) = g(x^2+y^2)$ for a func g

Suppose X, Y: cont. rand vars w/ diff'able PDFs f_X, f_Y; X, Y: indep; joint density function only depends on X^2+Y^2

Then, X, Y: normally distributed

<u>Proof</u>

We know $f_X(x) f_Y(y) \overset{\text{indep}}{=} f_{X,Y}(x,y) \overset{\text{last condition}}{=} g(x^2+y^2)$

Then, diff w.r.t. x: $f_X'(x) f_Y(y) = 2x g'(x^2+y^2)$

Rearranging, we get $\frac{f_X'(x)}{x f_X(x)} = \frac{2g'(x^2+y^2)}{g(x^2+y^2)}$

Claim: $\forall x_1, x_2$, $\frac{f_X'(x_1)}{x_1 f_X(x_1)} = \frac{f_X'(x_2)}{x_2 f_X(x_2)}$ (i.e. $\frac{f_X'(x)}{x f_X(x)} \ni$ const)

<u>Proof</u>: Choose y_1, y_2, s.t. $x_1^2 + y_1^2 = x_2^2 + y_2^2$. ✓

(Continue next time)